

Marketing Budget (In Thousands) vs Actual Sales (In Millions)

I. INTRO

The Dataset I have picked for data analysis is entitled Marketing Budget vs Actual Sales. It talks about how spending on marketing translates into actual sales into the company, and how it would be beneficial for the company. To show the computation attached here is the Google Sheets link [sales_dataset](#).

II. ANALYZING DATASET WITH STATISTICAL CONCEPTS

A. FREQUENCY DISTRIBUTION

CLASS INTERVAL	BIN	FREQUENCY	CM	CB	<CF	>CF	RF (%)
101.91 - 131.40	131.40	8	116.65	101.905 - 131.405	8	222	3.60
131.41 - 160.90	160.90	19	146.155	131.405 - 160.905	27	214	8.56
160.91 - 190.40	190.40	30	175.655	160.905 - 190.405	57	195	13.51
190.41 - 219.90	219.40	22	205.155	190.405 - 219.905	79	165	9.91
219.91 - 249.40	249.40	22	234.655	219.905 - 249.405	101	143	9.91
249.41 - 278.90	278.90	24	264.155	249.405 - 278.905	125	121	10.81
278.91 - 308.40	308.40	26	293.655	278.905 - 308.405	151	97	11.71
308.41 - 337.90	337.90	32	323.155	308.405 - 337.905	183	71	14.41
337.91 - 367.40	367.40	21	352.655	337.905 - 367.405	204	39	9.46
367.41 - 396.90	396.90	18	382.155	367.405 - 396.905	222	18	8.11

FINDINGS:

- The class intervals allow us to understand how actual sales (frequencies) change as marketing budgets rise. They also display the marketing budget, in thousands, that a company has allocated.
- The frequency part tells us that the greatest number of sales are produced by moderate marketing budgets like (308.41–337.90), indicating an ideal range for expenditure.
- Very few sales are attributed to budgets under 160.90, indicating that extremely low marketing expenditures are not enough to generate a good amount of sales.
- Although a budget from 308 - 337.90 is the optimal range, sometimes a cheaper budget, 160.91 - 190.40, for different campaigns can work as well. This tells us that marketing is a major factor in sales, but it does not tell us that it is the main factor of boost in sales and there are instances where a campaign may need to rely on different aspects to boost sales other than marketing.
- It also tells us that sales start to decline after spending more than 337.90 (337.91 – 367.40 and 367.41–396.90).
- When higher spending does not result in comparably higher sales, this could be a sign of declining returns.

B. CENTRAL TENDENCIES AND DISPERSION

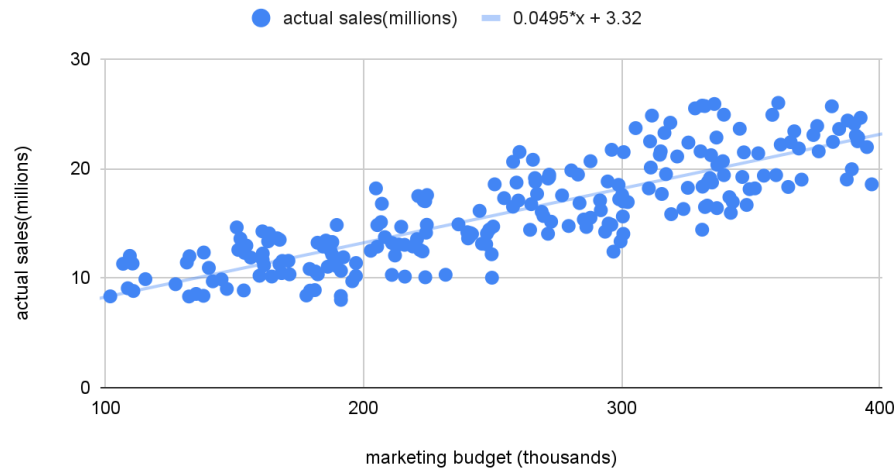
MIN	MAX	RANGE	MEAN	MEDIAN	MODE	VARIANCE
101.91	396.88	294.97	257.99	262.49	287.93	6152.73

FINDINGS:

- The range 294.97, indicating a broad spectrum of spending levels on marketing campaigns.
- The mean of the marketing budgets is 101.91 to 396.88. This large range indicates that marketing expenditures vary significantly. This can be the result of varying marketing sizes, audience targeting, or seasonal modifications.
- The average amount that businesses spend on marketing is 257.99
- 262.49 is the median marketing budget. The median shows that companies' spending is fairly evenly dispersed with little to no irregularity because it is near the mean.
- There is a significant disparity in the marketing budgets, as indicated by the high variance and standard deviation. This implies that the amount companies spend on various campaigns may vary. Although there are ways to strategically use variety (such as increased spending for bigger campaigns),

C. CORRELATION AND REGRESSION

actual sales(millions) vs. marketing budget (thousands)

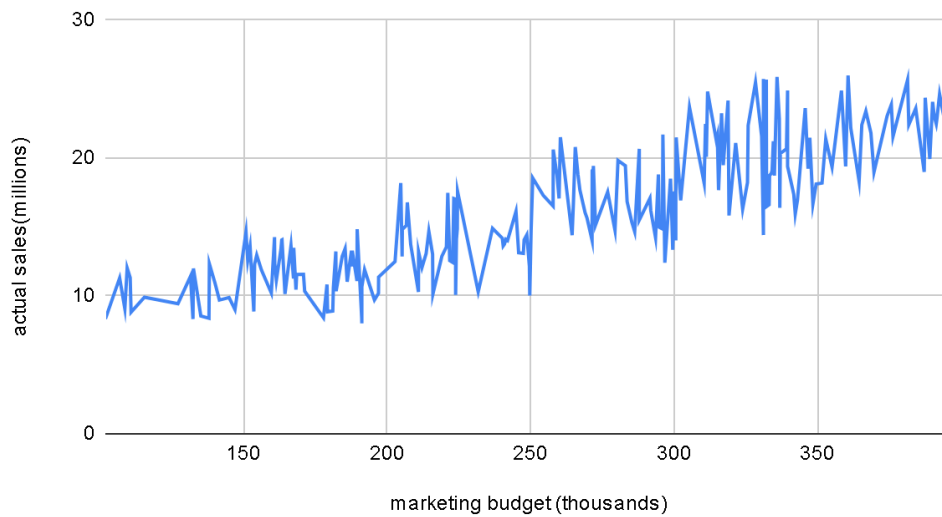


FINDINGS:

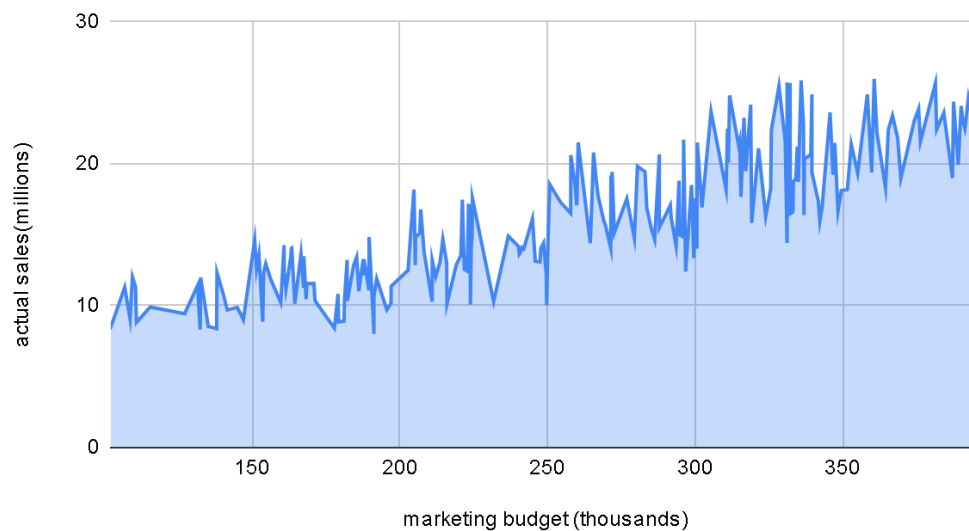
Presented to you right now is the scatter plot of the dataset, it shows that there is a high-positive correlation between the marketing budget and the results which are the actual sales. The range, mean, median, mode, variance, and standard deviation computed make this clear. better marketing spending is typically linked to better sales, as seen by the large range and high variability as well as the steady rise in average and median values. The regression equation tells us the following things:

1. Increasing the budget will most likely generate higher numbers in sales.
2. The equation allows us to predict or assume a possible number of sales once the budget for marketing is increased or decreased.
3. The variable y, which is 3.32 million sales, tells us that if no budget is allocated for marketing then the total sales could be near or be exactly 3.32 million worth of sales.

actual sales(millions) vs. marketing budget (thousands)



actual sales(millions) vs. marketing budget (thousands)



The line and area chart shown above further proves our findings. Based on the charts above, it further shows that our mean, around the range of 257.99 thousand is the optimal budget for marketing. As there are multiple factors that affect the sales, it shows that balance in budgeting is vital and focusing in different areas of the business is important as well in order to boost sales.

REGRESSION ANALYSIS:

Let X be the marketing budget (in thousands)

Let Y be the number of actual sales (in millions)

Equation:

$$Y = 0.0495x + 3.32$$

REGRESSION ANALYSIS		
CORRELATION:	Sample Independent Variable	Findings (Sales)
0.84	122.98	9.40751
Strong Positive Correlation	258.47	16.114265
	340.56	20.17772
	400	23.12
	450	25.595

The study emphasizes the role of marketing in driving sales growth. However, it is also vital to understand the principle of diminishing returns, which can apply when marketing costs grow excessively. While raising budgets generally boosts sales, the incremental sales gain per additional unit of budget may fade over time. This insight demonstrates that firms should strive to manage resources efficiently, maximizing returns on marketing campaigns while reducing wasteful spending.

Furthermore, the strong positive correlation implies that marketing budget allocation is a reliable predictor of sales performance. Other factors that may influence sales include product quality, market competition, and consumer behavior. These additional elements should be included in a comprehensive marketing plan.

III. CONCLUSION

According to the data presented, there is a favorable association between allocating a marketing budget and increasing revenue. However, marketing cost is not the only factor impacting sales performance. Other important elements to consider include product quality, after-sales support, and consumer impression. This is clear in the frequency table, which demonstrates that a smaller marketing budget (e.g., 160.91-190.40) was able to achieve roughly similar sales as greater budget allocations (308.41-337.90).

Furthermore, the data shows that the best marketing expenditure is around 257.99, as this range produced the highest sales. This shows that keeping budget allocation within the appropriate range can help reduce variability and enhance sales forecasting. Variable budgets may allow for greater flexibility in campaign design, but this might result in unpredictable sales results. As a result, keeping consistency in budget allocation within the appropriate range is critical for effective financial planning.

Furthermore, regression analysis and data insights emphasize the value of data-driven decision-making. Understanding the relationship between budget and sales allows firms to plan strategically and prevent overspending. While having an adequate budget for marketing is vital, overpaying or devoting too many resources to marketing does not always provide proportional results.

Finally, organizations should strive for a balance and sticking to the ideal budget while also increasing customer engagement and retention. An excellent product, along with an effective marketing plan that piques client interest, is crucial. Ultimately, the product is the most valuable asset. A well-thought-out marketing strategy may enhance this value and ensure long-term corporate success.

Citations:

Gillum, S. (2013). How marketing impacts sales performance. Forbes.

<https://www.forbes.com/sites/gyro/2013/02/25/how-marketing-impacts-sales-performance/>

Haryanto, T., & Retnaningrum, M. (2020). The impact of marketing expenditure on firm performance. *Jurnal Manajemen Bisnis*, 11(2), 188-201. <https://doi.org/10.18196/mb.11297>

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